Static Electricity

Charging by Contact and Induction

Challenge!

Using what we learned this week, your challenge is to make your pop can roll as FAST as possible.

Discuss with your group how you can make this happen. Materials available to you:

- Acetate stick
- Ebonite rod
- Fur
- Wool
- Paper towel

What was your strategy? Why did this work?

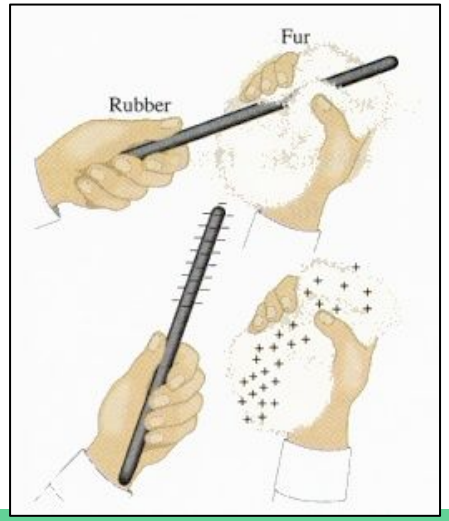
Let’s review what we’ve learned so far!

Recall: Objects can be charged in three ways

1 - Charge by Friction

When 2 <u>different</u> *neutral* objects are rubbed together and electric charges are transferred from 1 object to the other

- One material is more likely to attract e^- and one more likely to give up
- Electrostatic Series



Recall: Electrostatic series

- Tells us how strongly a material holds onto its electrons
- Rubbing two objects together can sometimes cause electrons to transfer from one object to another, leaving them charged
- Objects that lost their electrons become *positive*
- Objects that gain electrons become *negative*

Table 1 Electrostatic Series

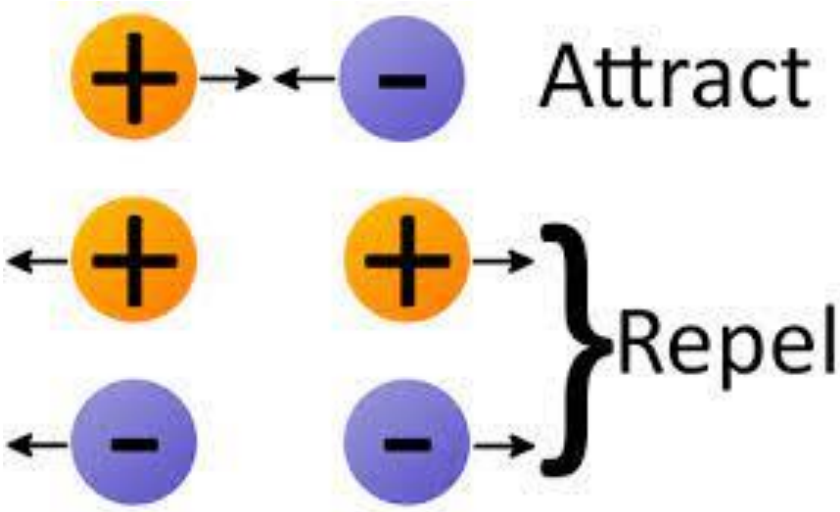
Material	Charge tendency
human skin	+
rabbit fur	(weaker tendency to gain electrons)
acetate	
glass	
human hair	
nylon	
wool	
cat fur	
silk	
paper	
cotton	
wood	
amber	
rubber balloon	(stronger tendency to gain electrons)
vinyl	
polyester	
ebonite	−

Recall: Law of Electric Charges

1 - Opposites attract

2 - Like charges repel

3 - Neutral object + charged object → induced charge separation (polarized object)



How do we know when an object is charged?

Electroscope : instrument used to detect electric charges

Pith ball electroscope - ball of pith (plant material) suspended from a stand by a thread

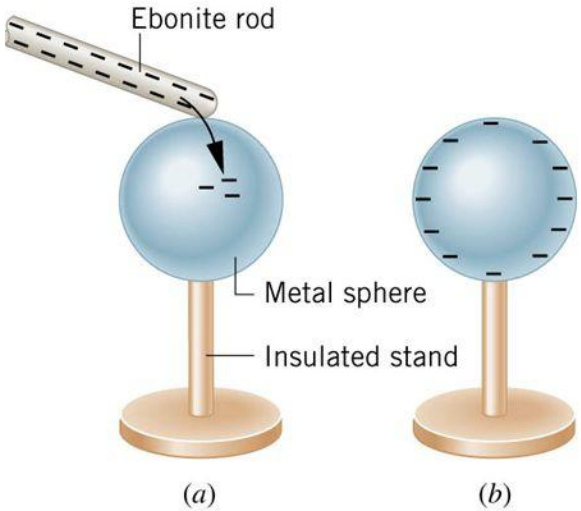
- Bring charged object near; if it is charged, the ball will move

Gold leaf electroscope - more sensitive

- Leaves separate when charged

2 - Charge by Conduction (*contact*)

- When objects with different amounts of **electric charge** come in contact, electrons will move from the object with more **negative charge** , to the object with smaller **negative charge**
- Charging by contact alway harge in both objects

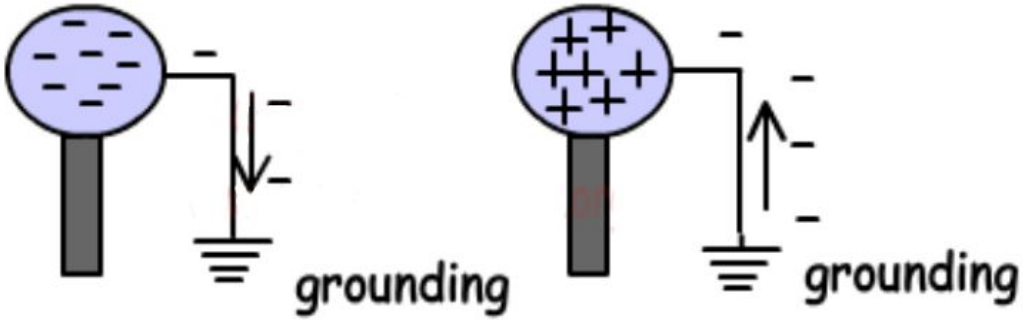


Grounding

Connecting an object to a large body, like **Earth** , that is capable of effectively removing an electric charge that the object might have.

Charged objects, either **+ or -** , can have their excess charge removed by grounding

Large objects can '**neutralize** ' because there is infinite room for electrons



3 - Charging by Induction

Simulation

Charging **without** contact. *Always results in **OPPOSITE charges** *

Two types:

Temporary

- Bringing a charged object toward a neutral will cause an **induced charge separation**

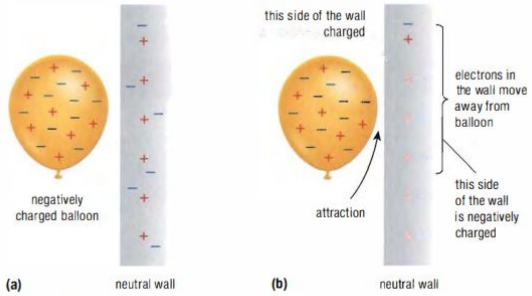
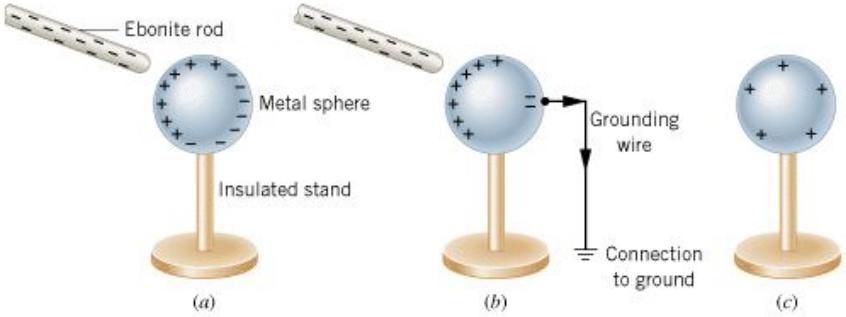


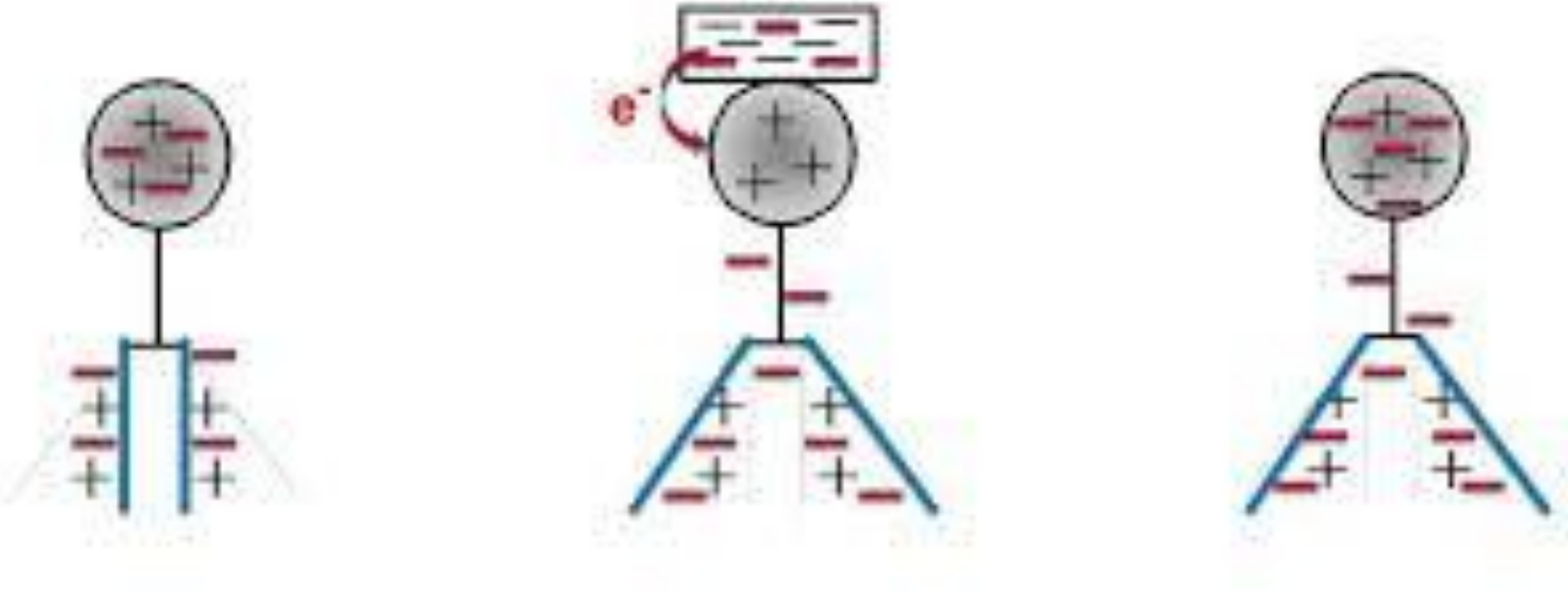
Figure 1 A negatively charged balloon (a) is brought near a neutral wall, causing an induced charge separation in the wall. (b) The wall becomes temporarily charged by induction.

Permanent

- Objects can be permanently charged by induction if **grounded**

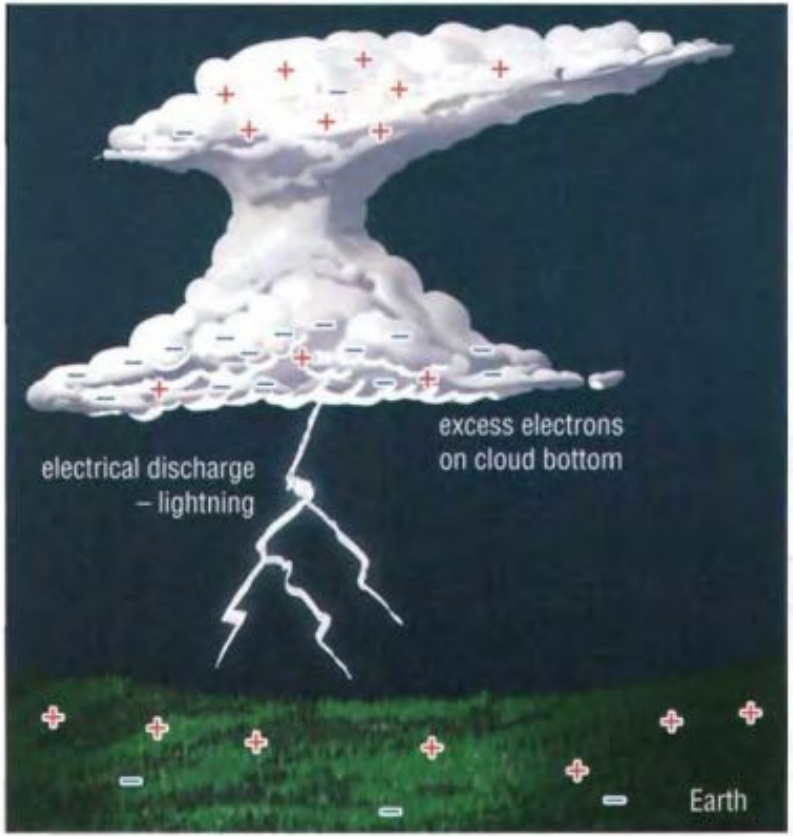


Charging by Contact

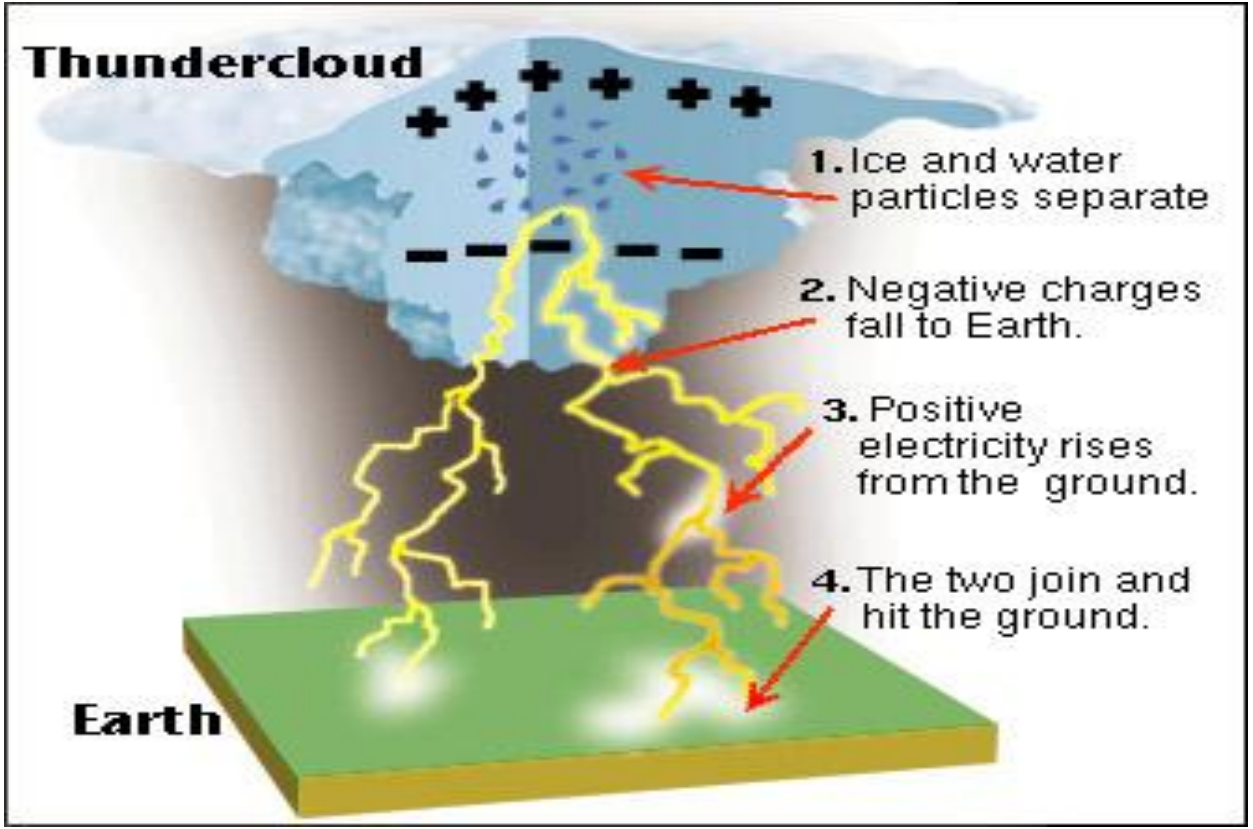


Lightning

- Theory is that as **water droplets** in clouds move past one another, they become charged
- Negatively charged water molecules collect at the bottom of clouds
- This **repels electrons** at Earth's surface, resulting in positive charge (charge imbalance)
- When this **charge imbalance** becomes great enough, excess **electrons** rapidly transfer from the cloud to the ground in the form of lightning



LIGHTNING


Thundercloud
1. Ice and water particles separate
2. Negative charges fall to Earth.
3. Positive electricity rises from the ground.
4. The two join and hit the ground.
Earth

BOOM!
WHAT IS
THUNDER?

Your turn!

- Charging by Contact and Induction Practice